

## 1. Design of CMOS Inverter

**AIM:** To design, simulate and verify the operation of CMOS inverter using LT Spice tool. Verify the following 1) DC Analysis 2) Transient Analysis

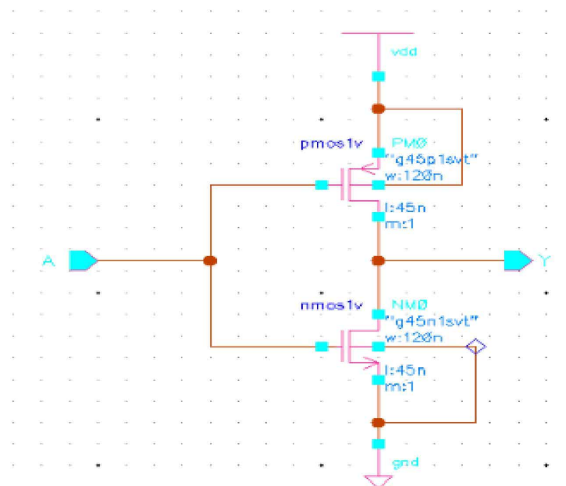
### Tools used:

1. LT Spice Tool for Schematic Design
2. And for Simulation

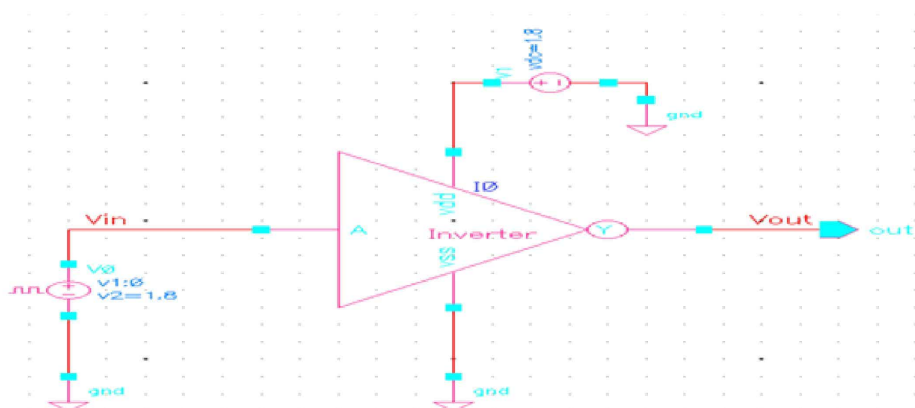
### Procedure:

- Open the LT Spice tool, click on file ➡ new Schematic
- Select the components from the palate and place it on the window.
- Complete the design and right click on voltage source add DC value.
- Click on the configure analysis, select DC sweep for DC analysis and observe the O/P.
- Next right click on the voltage source select ADVANCE add values.
- Next click on configure analysis, select TRANSIENT add all parameters, and then click on OK.
- Observe the Transient Analysis.

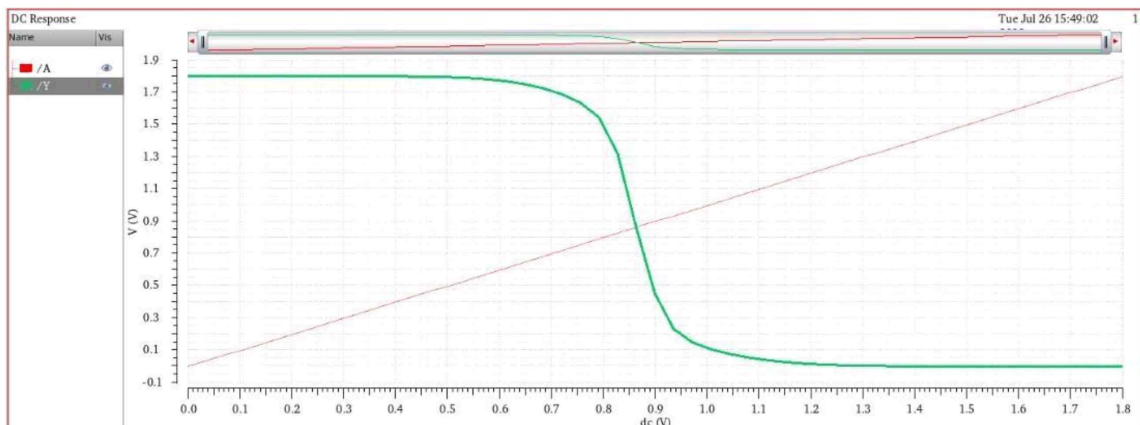
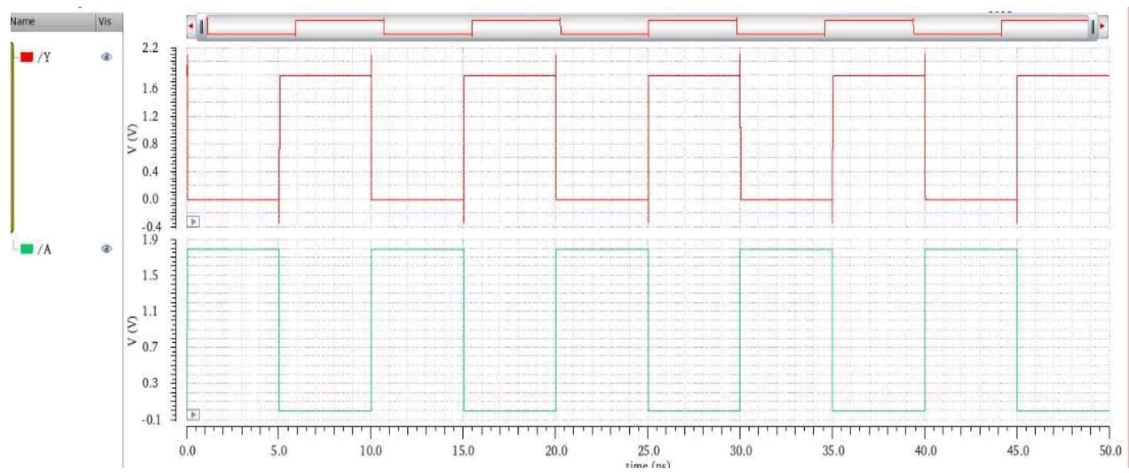
### Circuit:



### Testbench:



## Simulation Results:



Transient and DC analysis

## Result:

The design, simulation and verification of DC Analysis and Transient Analysis for CMOS inverter using LT Spice tools was performed.

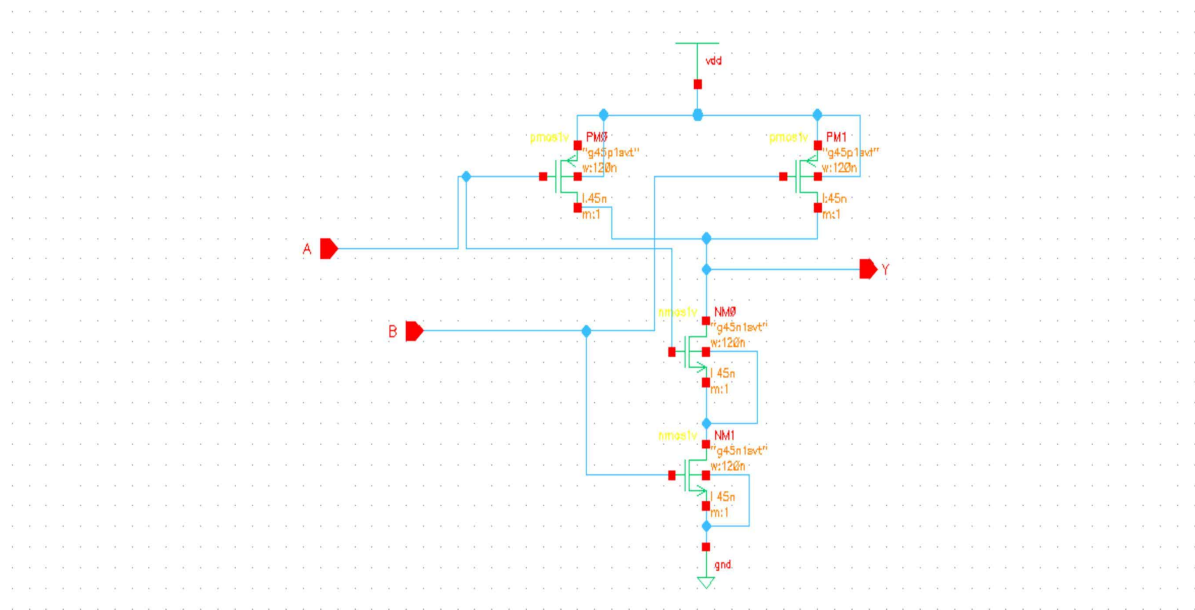
## 2. Design of 2-input NAND Gate Design

**AIM:** To design, simulate and verify the operation of CMOS NAND using LT Spice tool. Verify the following 1) DC Analysis 2) Transient Analysis

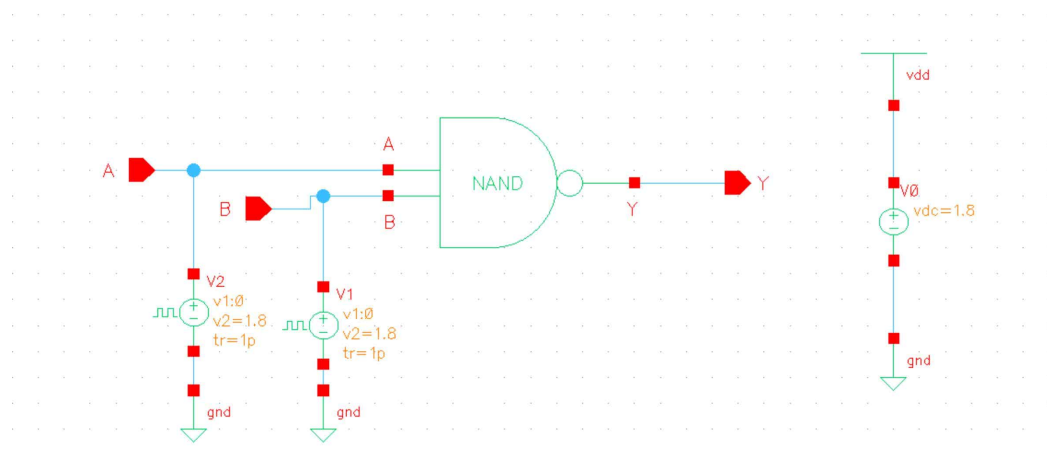
Tools used:

1. LT Spice Tool for Schematic Design
2. And for Simulation

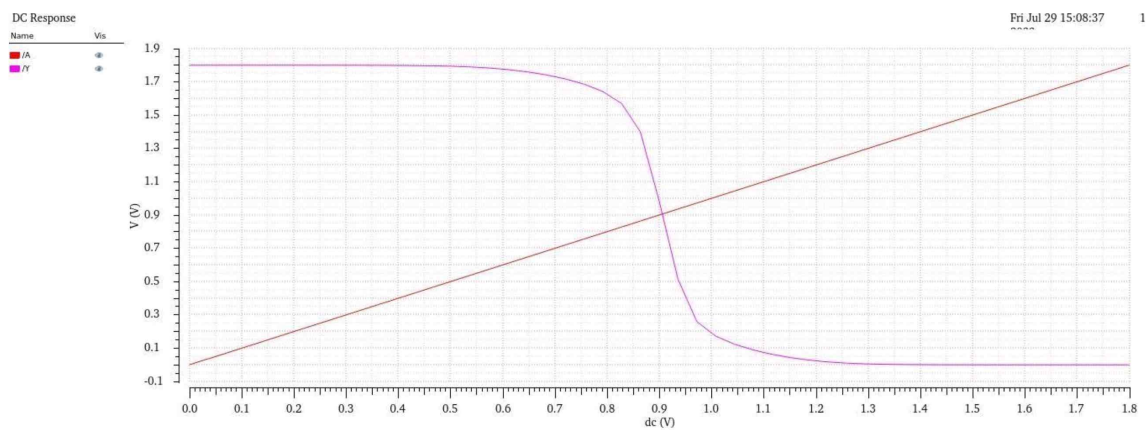
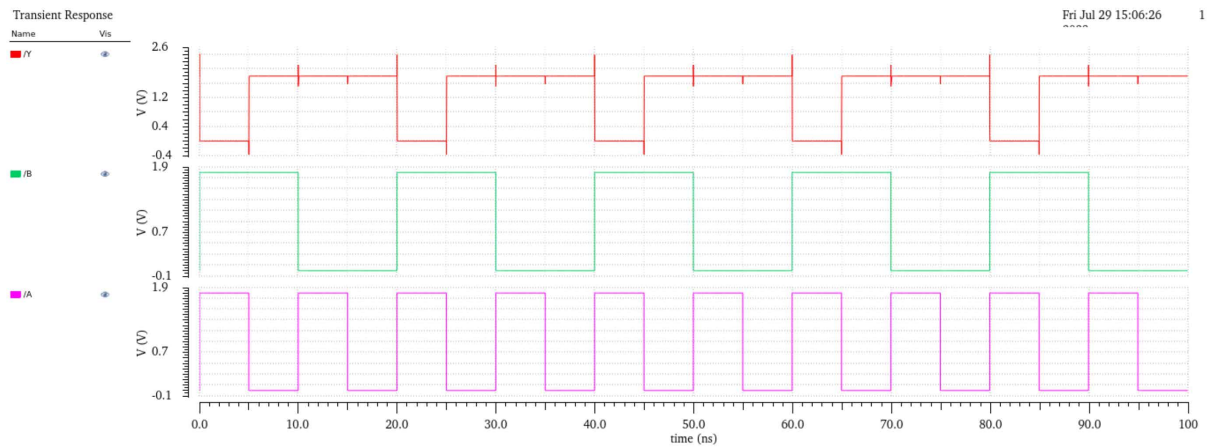
Circuit:



Testbench:



## Simulation Results:



## Transient and DC analysis

### Result:

The design, simulation and verification of DC Analysis and Transient Analysis for CMOS NAND gate using LT Spice tool was performed.

### 3. DESIGN COMMON SOURCE AMPLIFIER

AIM:

To perform the Design and Simulation of Common Source Amplifier circuit through schematic entry.

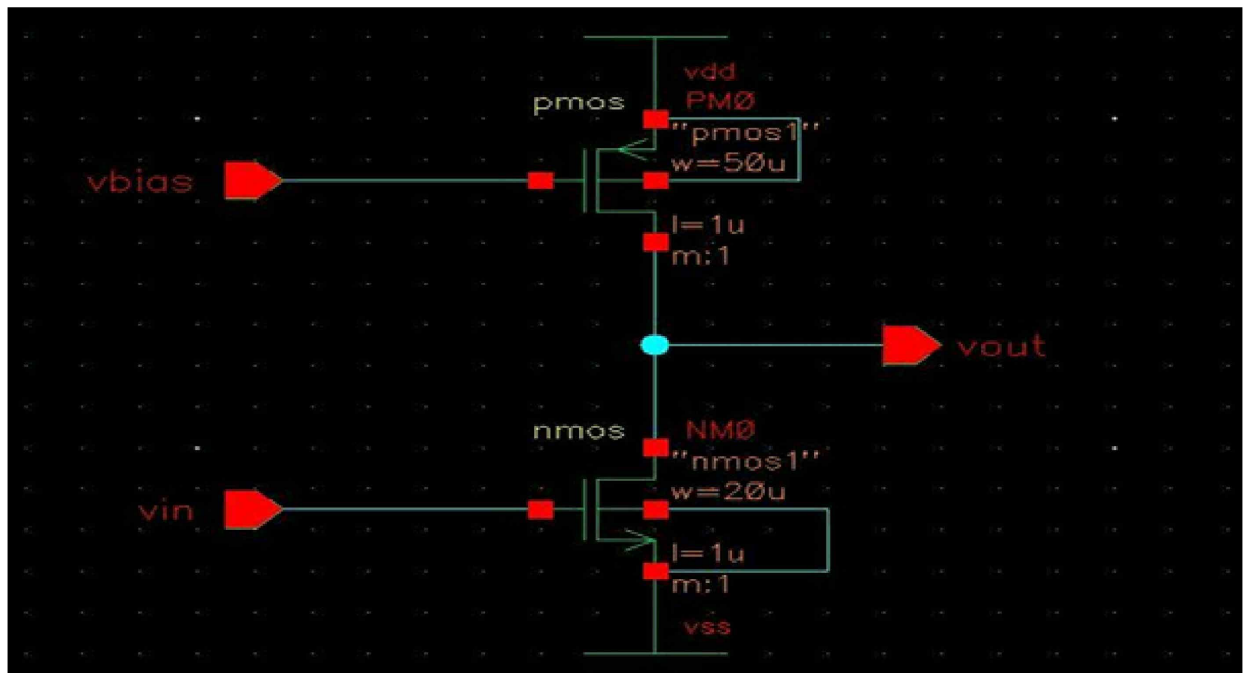
Tools used:

1. LT Spice Tool for Schematic Design
2. And for Simulation

**Procedure:**

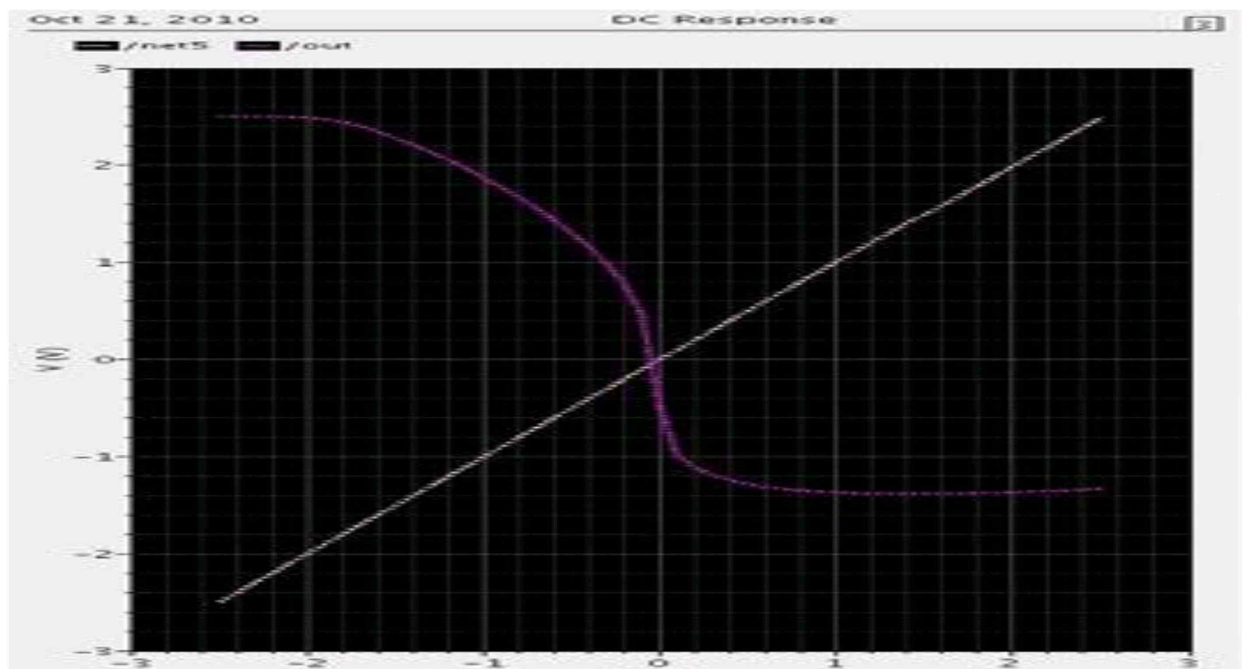
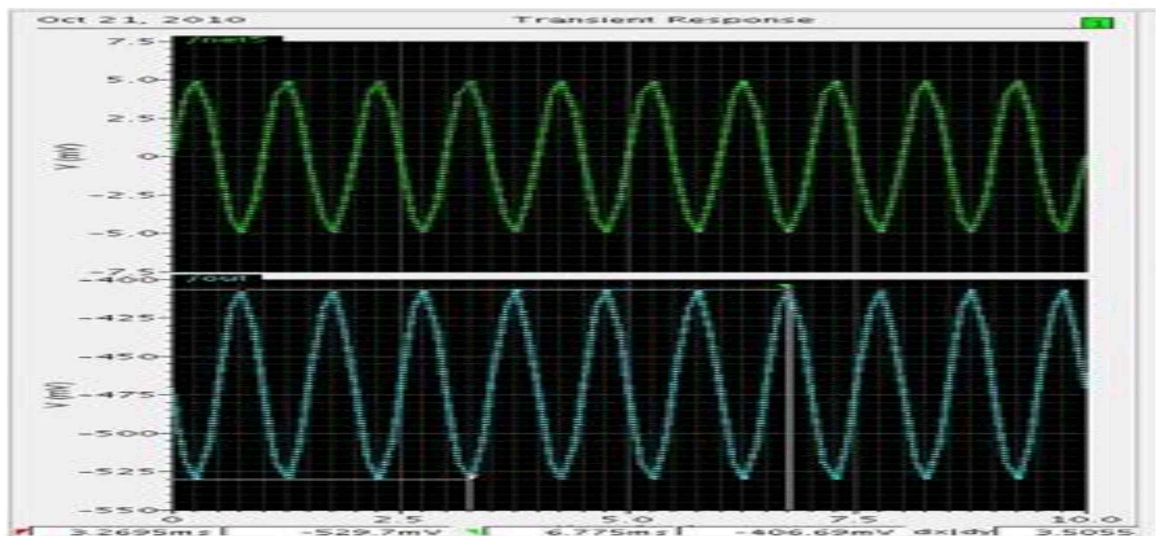
- Open the LT Spice tool, click on file → new Schematic
- Select the components from the palette and place it on the window.
- Complete the design and right click on voltage source add DC value.
- Click on the configure analysis, select DC sweep for DC analysis and observe the O/P.
- Next right click on the voltage source select ADVANCE add values.
- Next click on configure analysis, select TRANSIENT add all parameters, and then click on OK.
- Observe the Transient Analysis.

Circuit:



COMMON SOURCE AMPLIFIER

Simulation Results:



Transient and DC analysis

Result:

The design, simulation and verification of DC Analysis and Transient Analysis for Common source amplifier using LT Spice tool was performed.



#### 4. DESIGN COMMON DRAIN AMPLIFIER

AIM:

To perform the Design and Simulation of Common drain amplifier circuit through schematic entry.

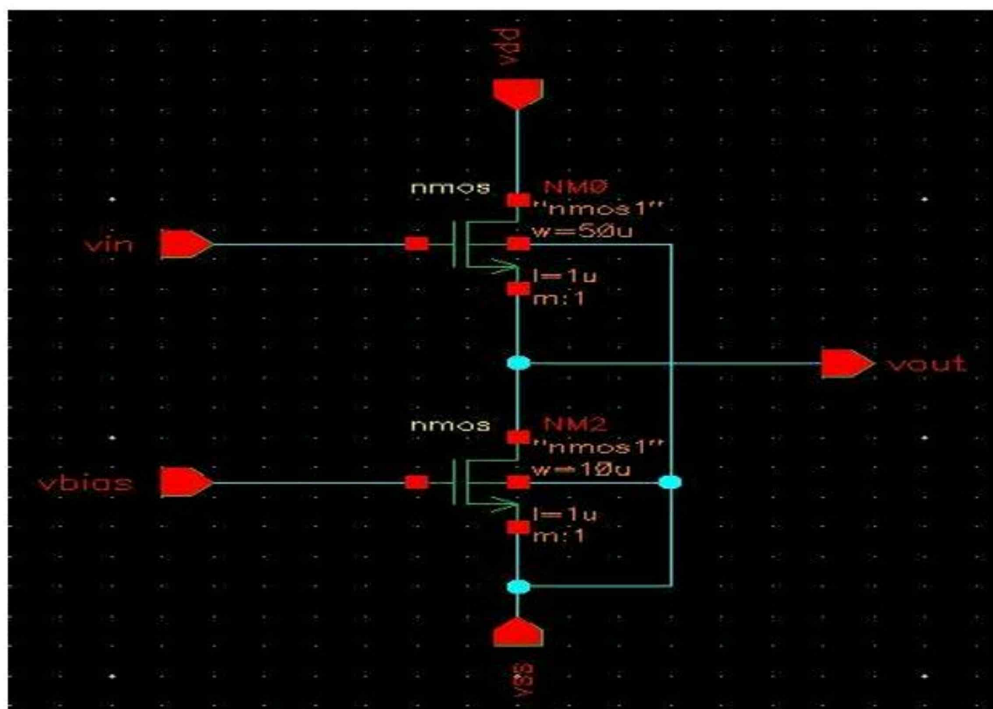
Tools used:

1. LT Spice Tool for Schematic Design
2. And for Simulation

**Procedure:**

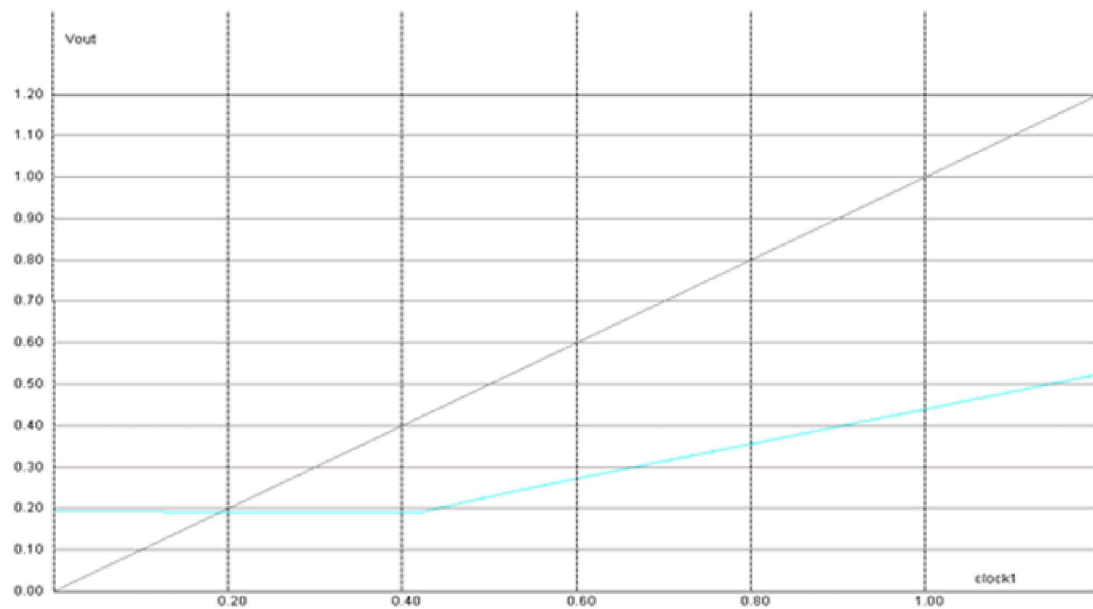
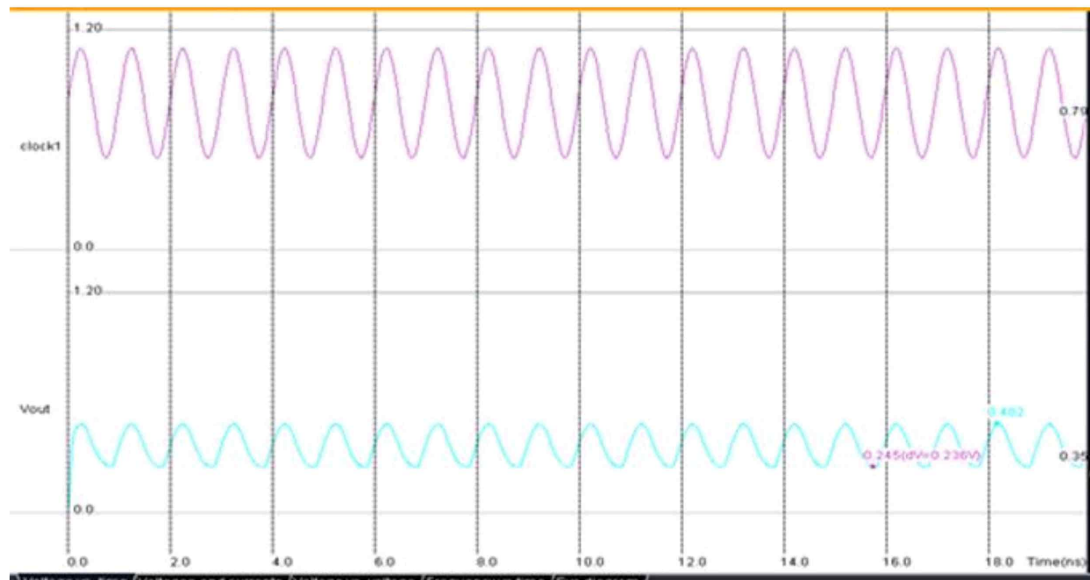
- Open the LT Spice tool, click on file → new Schematic
- Select the components from the palate and place it on the window.
- Complete the design and right click on voltage source add DC value.
- Click on the configure analysis, select DC sweep for DC analysis and observe the O/P.
- Next right click on the voltage source select ADVANCE add values.
- Next click on configure analysis, select TRANSIENT add all parameters, and then click on OK.
- Observe the Transient Analysis.

Circuit:



COMMON DRAIN AMPLIFIER

## Simulation Results:



## Transient and DC analysis

## Result:

The design, simulation and verification of DC Analysis and Transient Analysis for Common drain amplifier using LT Spice tool was performed.